

PRACTICAL NO 4

AIM:- Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE

```
import hashlib
import tkinter as tk
from tkinter import messagebox

# ----- RSA KEY GENERATION -----

def generate_keys():
    p = 61
    q = 53

    n = p * q
    phi = (p - 1) * (q - 1)

    e = 17
    d = 2753

    public_key = (e, n)
    private_key = (d, n)

    return public_key, private_key

# ----- HASH FUNCTION -----

def create_hash(message):
    hash_value = hashlib.sha256(message.encode()).hexdigest()
    return int(hash_value, 16)

# ----- CREATE DIGITAL SIGNATURE -----

def create_signature(message, private_key):
    d, n = private_key

    hash_value = create_hash(message)

    signature = pow(hash_value, d, n)

    return signature

# ----- VERIFY SIGNATURE -----
```

```

def verify_signature(message, signature, public_key):
    e, n = public_key

    hash_value = create_hash(message)

    decrypted_signature = pow(signature, e, n)

    return decrypted_signature == hash_value % n

# ----- CLI MODE -----

def cli_program():

    public_key, private_key = generate_keys()

    while True:

        print("\n=====")
        print("    RSA DIGITAL SIGNATURE")
        print("=====")
        print("1. Generate Digital Signature")
        print("2. Verify Digital Signature")
        print("3. Show RSA Keys")
        print("4. Exit")

        choice = input("\nEnter your choice: ")

        # Generate Signature
        if choice == "1":

            message = input("\nEnter message: ")

            signature = create_signature(
                message,
                private_key
            )

            print("\nMessage:", message)
            print("Digital Signature:", signature)

        # Verify Signature
        elif choice == "2":

            message = input("\nEnter message: ")
            signature_input = input("Enter digital signature: ")

            try:
                signature = int(signature_input)

                result = verify_signature(

```

```

        message,
        signature,
        public_key
    )

    if result:
        print("\n✓ Signature Verified")
        print("✓ Message is authentic and unchanged.")

    else:
        print("\nX Signature Verification Failed")
        print("X Message may have been modified.")

    except ValueError:
        print("\nInvalid signature. Enter numbers only.")

# Show Keys
elif choice == "3":

    print("\nPublic Key :", public_key)
    print("Private Key:", private_key)

# Exit
elif choice == "4":

    print("\nExiting program...")
    break

else:

    print("\nInvalid choice. Please try again.")

# ----- GUI FUNCTIONS -----

def gui_generate():

    message = message_entry.get()

    if message == "":
        messagebox.showwarning(
            "Warning",
            "Please enter a message."
        )
    return

    signature = create_signature(
        message,
        private_key
    )

```

```

signature_entry.delete(0, tk.END)
signature_entry.insert(0, str(signature))

result_label.config(
    text="Digital signature generated successfully."
)

def gui_verify():

    message = message_entry.get()
    signature_text = signature_entry.get()

    if message == "":
        messagebox.showwarning(
            "Warning",
            "Please enter a message."
        )
        return

    if signature_text == "":
        messagebox.showwarning(
            "Warning",
            "Please enter a digital signature."
        )
        return

    try:

        signature = int(signature_text)

        result = verify_signature(
            message,
            signature,
            public_key
        )

        if result:

            result_label.config(
                text="✓ Signature Verified\n"
                "✓ Message is authentic and unchanged."
            )

        else:

            result_label.config(
                text="X Verification Failed\n"
                "X Message has been modified."
            )

```

```
except ValueError:
```

```
    messagebox.showerror(
        "Error",
        "Digital signature must contain numbers only."
    )
```

```
# ----- GUI MODE -----
```

```
def start_gui():
```

```
    global message_entry
    global signature_entry
    global result_label
```

```
    root = tk.Tk()
```

```
    root.title("RSA Digital Signature")
    root.geometry("550x450")
```

```
    title = tk.Label(
        root,
        text="RSA DIGITAL SIGNATURE",
        font=("Arial", 18, "bold")
    )
```

```
    title.pack(pady=20)
```

```
    tk.Label(
        root,
        text="Enter Message:"
    ).pack()
```

```
    message_entry = tk.Entry(
        root,
        width=55
    )
```

```
    message_entry.pack(pady=8)
```

```
    tk.Button(
        root,
        text="Generate Digital Signature",
        command=gui_generate,
        width=30
    ).pack(pady=10)
```

```
    tk.Label(
        root,
        text="Digital Signature:"
    )
```

```

).pack()

signature_entry = tk.Entry(
    root,
    width=55
)

signature_entry.pack(pady=8)

tk.Button(
    root,
    text="Verify Signature",
    command=gui_verify,
    width=30
).pack(pady=10)

result_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)

result_label.pack(pady=20)

tk.Label(
    root,
    text="Public Key: " + str(public_key)
).pack()

root.mainloop()

# ----- MAIN PROGRAM -----

public_key, private_key = generate_keys()

while True:

    print("\n=====")
    print("  RSA DIGITAL SIGNATURE SYSTEM")
    print("=====")
    print("1. CLI Mode")
    print("2. GUI Mode")
    print("3. Show Keys")
    print("4. Exit")

    choice = input("\nEnter your choice: ")

    if choice == "1":

        cli_program()

```

```

elif choice == "2":

    start_gui()

elif choice == "3":

    print("\nPublic Key :", public_key)
    print("Private Key:", private_key)

elif choice == "4":

    print("\nProgram terminated.")
    break

else:

    print("\nInvalid choice. Please try again.")

```

OUTPUT:-

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/as993/Downloads/T118 CS PRACTICAL NO 4.py =====

=====
RSA DIGITAL SIGNATURE SYSTEM
=====
1. CLI Mode
2. GUI Mode
3. Show Keys
4. Exit
Enter your choice: 1

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit
Enter your choice: 1
Enter message: HELLO ABHISHEK
Message: HELLO ABHISHEK
Digital Signature: 2584

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

```

```
*IDLE Shell 3.14.3*
File Edit Shell Debug Options Window Help

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

Enter your choice: 2

Enter message: HELLO ABHISHEK
Enter digital signature: 2584

✓ Signature Verified
✓ Message is authentic and unchanged.

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

Enter your choice: 3

Public Key : (17, 3233)
Private Key: (2753, 3233)

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

Enter your choice:
```

```
*IDLE Shell 3.14.3*
File Edit Shell Debug Options Window Help

✓ Signature Verified
✓ Message is authentic and unchanged.

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

Enter your choice: 3

Public Key : (17, 3233)
Private Key: (2753, 3233)

=====
RSA DIGITAL SIGNATURE
=====
1. Generate Digital Signature
2. Verify Digital Signature
3. Show RSA Keys
4. Exit

Enter your choice: 4

Exiting program...

=====
RSA DIGITAL SIGNATURE SYSTEM
=====
1. CLI Mode
2. GUI Mode
3. Show Keys
4. Exit

Enter your choice: 2
```

RSA Digital Signature

Enter Message:

HELLO ABHISHEK

Generate Digital Signature

Digital Signature:

2584

Verify Signature

Digital signature generated successfully.

Public Key: (17, 3233)

